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FS 0430

Preguntas conceptuales

1. No. Ambas fuerzas, la gravitacional y la eléctrica, son $\propto \frac{1}{d^2}$, entonces no se gana nada variando d.
Matemáticamente:

$$\frac{F_e}{F_g} = 2,3 \times 10^{39}$$

$$\Leftrightarrow \frac{\left(\frac{K_e q_1 q_2}{\cancel{d^2}}\right)}{\left(\frac{G m_1 m_2}{\cancel{d^2}}\right)} = 2,3 \times 10^{39}$$

$$\Leftrightarrow \frac{K_e q_1 q_2}{G m_1 m_2} = 2,3 \times 10^{39}$$

No hay dependencia de d.

2. 1. La fuerza es de repulsión.

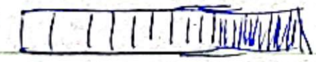
2. Por 3era Ley de Newton, las fuerzas experimentadas son igual en magnitud.

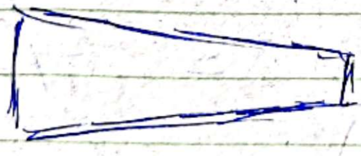
Entonces la opción e

3. La fuente del campo eléctrico

4. Ley de Gauss es útil para ~~sig~~ calcular el campo eléctrico para distribuciones de carga simétricas.

$\sigma(x) \propto x$.

a.  barra con densidad de carga $\sigma(x) \propto x$.

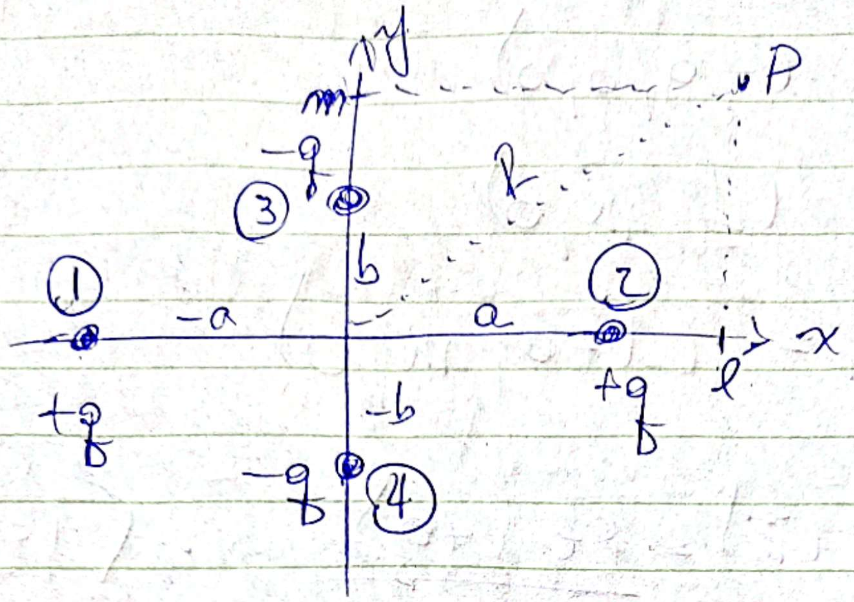
b.  barra con densidad de carga uniforme, pero asimétrica.

c.  Media esfera

En todos estos casos el $\vec{E}$ no es uniforme para superficies gaussianas

Problema #1

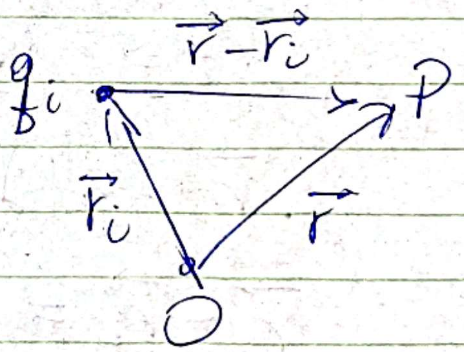
(a)



$R = \sqrt{l^2+m^2}$

$$\vec{E}_i(\vec{r}) = K_e \frac{q_i}{|\vec{r}-\vec{r}_i|^3}(\vec{r}-\vec{r}_i)$$

$$\vec{E}(\vec{r}) = K_e \sum_{i=1}^{4} \frac{q_i\,(\vec{r}-\vec{r}_i)}{|\vec{r}-\vec{r}_i|^3}$$



donde

$\vec{r}$: vector posición de P

$\vec{r}_i$: vector posición de q_i

$\vec{r}-\vec{r}_i$: vector de $q_i \to P$.

Para ①.

$\vec{r}_1 = (-a, 0, 0)$

$\vec{r} = (l, m, 0)$

$\vec{r} - \vec{r}_1 = (l+a, m, 0)$

$$\vec{E}_1(\vec{r}) = \frac{k_e(+q)}{[(l+a)^2 + m^2]^{3/2}} \cdot \langle l+a, m, 0 \rangle$$

Para ②:

$\vec{r}_2 = (a, 0, 0)$

$\vec{r} = (l, m, 0)$

$\vec{r} - \vec{r}_2 = (l-a, m, 0)$

$$\vec{E}_2(\vec{r}) = \frac{k_e \cdot (+q)}{[(l-a)^2 + m^2]^{3/2}} \cdot \langle l-a, m, 0 \rangle$$

Para ③:

$\vec{r_3} = (0, b, 0)$

$\vec{r} = (l, m, 0)$

$\vec{r} - \vec{r_3} = (l, m-b, 0)$

$$\vec{E}_3(\vec{r}) = \frac{K_e(-q)}{[l^2 + (m-b)^2]^{3/2}} \cdot \langle l, m-b, 0 \rangle$$

Para ④:

$\vec{r_4} = (0, -b, 0)$

$\vec{r} = (l, m, 0)$

$\vec{r} - \vec{r_4} = (l, m+b, 0)$

$$\vec{E}_4(\vec{r}) = \frac{K_e(-q)}{[l^2 + (m+b)^2]^{3/2}} \cdot \langle l, m+b, 0 \rangle$$

$$\vec{E}(\vec{r})_x = \vec{E}_1(\vec{r})_x + \vec{E}_2(\vec{r})_x + \vec{E}_3(\vec{r})_x + \vec{E}_4(\vec{r})_x$$

$$\vec{E}(\vec{r})_x = \frac{q}{4\pi\varepsilon_0}\left[\frac{l+a}{[(l+a)^2+m^2]^{3/2}} + \frac{l-a}{[(l-a)^2+m^2]^{3/2}} - \frac{l}{[l^2+(m-b)^2]^{3/2}} - \frac{l}{[l^2+(m+b)^2]^{3/2}}\right] \hat{i}$$

$$\vec{E}(\vec{r})_y = \vec{E}_1(\vec{r})_y + \vec{E}_2(\vec{r})_y + \vec{E}_3(\vec{r})_y + \vec{E}_4(\vec{r})_y$$

$$\vec{E}(\vec{r})_y = \frac{q}{4\pi\varepsilon_0}\left[\frac{m}{[(l+a)^2+m^2]^{3/2}} + \frac{m}{[(l-a)^2+m^2]^{3/2}}\right.$$

$$\left. - \frac{(m-b)}{[l^2+(m-b)^2]^{3/2}} - \frac{(m+b)}{[l^2+(m+b)^2]^{3/2}}\right]\hat{j}$$

$$\vec{E}(\vec{r})_z = 0\,\hat{k}$$

(b)

Está todo.

(C) para $m=0$

$$E(r)_x = Kq\left(\frac{l+a}{((l+a)^2)^{3/2}} + \frac{l-a}{[(l-a)^2]^{3/2}} - \frac{l}{(l^2+b^2)^{3/2}} - \frac{l}{(l^2+b^2)^{3/2}}\right)$$

$$E(r)_x = Kq\left(\underbrace{\frac{1}{(l+a)^2}}_{(*)} + \underbrace{\frac{1}{(l-a)^2}}_{(**)} - \underbrace{\frac{2l}{(l^2+b^2)^{3/2}}}_{(***)}\right)$$

$$R = \sqrt{l^2+m^2} \rightarrow R = \sqrt{l^2}$$

$$\rightarrow R = l$$

(*) $$\frac{1}{(l+a)^2} = \frac{1}{\left(l\left(1+\frac{a}{l}\right)\right)^2} = l^{-2}\left(1+\frac{a}{l}\right)^{-2}$$

$$l^{-2}\left(1+\frac{a}{l}\right)^{-2} = l^{-2}\left(1-\frac{2a}{l}+\frac{3a^2}{l^2}\ldots\right)$$

$$= \frac{1}{l^2} - \frac{2a}{l^3} + \frac{3a^2}{l^4}\ldots$$

(**) $$\frac{1}{(l-a)^2} = l^{-2}\left(1-\frac{a}{l}\right)^{-2}$$

$$= l^{-2}\left(1+\frac{2a}{l}+\frac{3a^2}{l^2}\ldots\right)$$

$$= \left(\frac{1}{l^2}+\frac{2a}{l^3}+\frac{3a^2}{l^4}\right)$$

$\circledast\circledast\circledast$: $\dfrac{\ell}{(\ell^2+b^2)^{3/2}} = \dfrac{\ell}{\ell^3\left(1+\frac{b^2}{\ell^2}\right)^{3/2}} = \dfrac{1}{\ell^2}\left(1+\frac{b^2}{\ell^2}\right)^{-3/2}$

$$= \frac{1}{\ell^2}\left(1 - \frac{3}{2}\frac{b^2}{\ell^2} + \cdots\right)$$

$$= \frac{1}{\ell^2} - \frac{3}{2}\frac{b^2}{\ell^4} + \cdots$$

Entonces $\circledast + \circledast\circledast + -2\circledast\circledast\circledast$

$$\frac{1}{\ell^2} - \frac{2a}{\ell^3} + \frac{3a^2}{\ell^4}$$

$$+\ \frac{1}{\ell^2} + \frac{2a}{\ell^3} + \frac{3a^2}{\ell^4}$$

$$-\ \frac{2}{\ell^2} + 3\frac{b^2}{\ell^4}$$

$$0 + 0 + \frac{3a^2+3a^2+3b^2}{\ell^4} = \frac{3(2a^2+b^2)}{\ell^4}$$

$\bar{E}(m=0)_x = Kq\,\dfrac{3(2a^2+b^2)}{\ell^4}$, Lo mismo para $E(m=0)_y$.

Problema #2

(a)

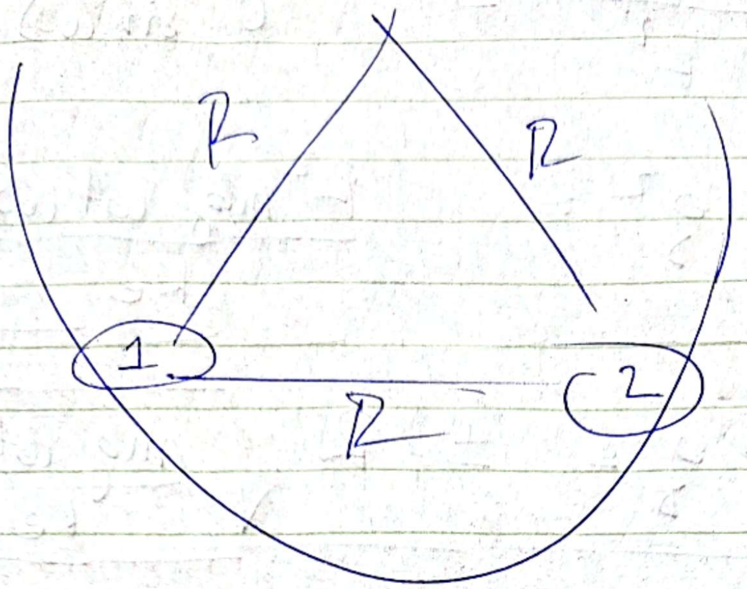
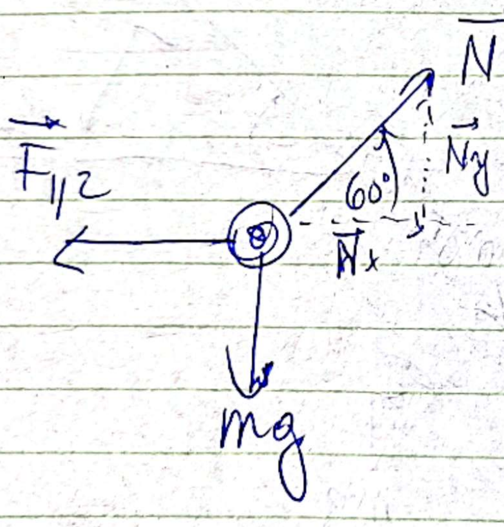


Diagrama de cuerpo libre para ①:



$$\begin{cases} \sum \vec{F}_x : \vec{N}_x - \vec{F}_{1,2} = \vec{0} & (1) \\ \sum \vec{F}_y : \vec{N}_y - m\vec{g} = \vec{0} & (2) \end{cases}$$

De ②:

$$N_y = mg$$

$$(\Rightarrow) \; N \sin(60) = mg$$

$$N = \frac{mg}{\sin(60)} \quad (3)$$

De ①:

$$\vec{F}_{1,2} = \vec{N}_x$$

$$\frac{K_e q^2}{R^2} = \cos(60) N \quad (4)$$

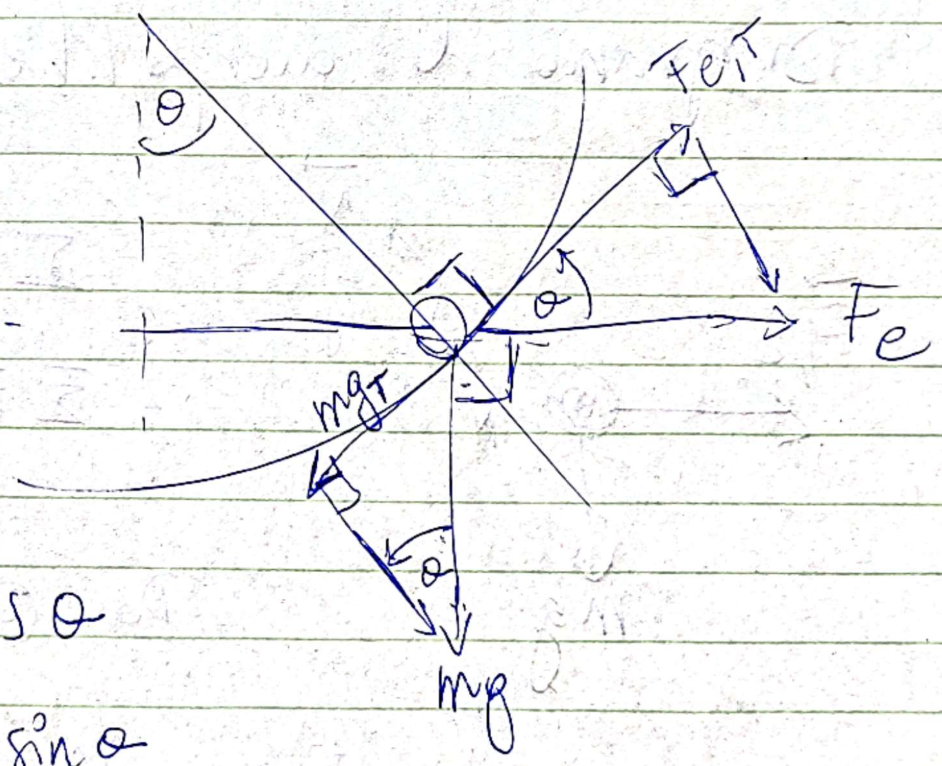
(3) en (4):

$$\frac{K_e q^2}{R^2} = mg \frac{\cos(60)}{\sin(60)}.$$

$$\Leftrightarrow \quad q^2 = \frac{R^2 mg \cot(60)}{K_e}$$

$$\Leftrightarrow \quad \boxed{q = \pm R \sqrt{\frac{mg \cot(60)}{K_e}}}$$

(b)



$F_{e,T} = F_e \cos\theta$

$mg_T = mg \sin\theta$

$$\Sigma F_T = m a_T$$

$$F_e \cos\theta - mg \sin\theta = m R \ddot{\theta}$$

Y ya de aquí está rudo seguir...

Problema # 3.

De Gauß: $A_{esf} = 4\pi r^2$

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\varepsilon_0}$$

$$\Rightarrow E(4\pi r^2) = \frac{Q_{enc}}{\varepsilon_0}$$

$$\Leftrightarrow E(\vec{r}) = \frac{1}{4\pi\varepsilon_0} \frac{Q_{enc}}{r^2}$$

$$\rho = \frac{Q_{enc}}{V} \longrightarrow Q_{enc} = \rho V$$

$$dQ_{enc}(r') = \rho \, dV$$

$$Q_{enc}(r') = \int_V \rho \, dV$$

$$Q_{enc}(r') = \int_0^{\pi}\int_0^{2\pi}\int_0^{r'} \frac{q}{\pi a^3} e^{-\frac{2r}{a}} r'^2 \sin\theta \, dr' \, d\phi \, d\theta$$

$$\int_0^{\pi}\int_0^{2\pi} \sin\theta \, d\phi \, d\theta$$

$$\uparrow$$

$$Q_{enc}(r') = \int_0^{r'} \frac{q}{\pi a^3} e^{-\frac{2r'}{a}} r'^2 \, dr' \cdot 4\pi$$

$$q_{enc}(r') = \frac{4\cancel{\pi}\, q}{\cancel{\pi} a^3} \underbrace{\int_0^{r'} e^{-2r'/a}\, r'^2\, dr'}_{I_1}$$

ahora hay que integrar por partes.

$$u = r'^2 \qquad v = -\frac{a}{2} e^{-\frac{2r'}{a}}$$

$$du = 2r'\, dr' \qquad dv = e^{-2r'/a}\, dr'$$

$$I_1 = uv - \int v\, du$$

$$I_1 = r'^2 \cdot \frac{(-a)}{2} e^{-2r'/a} + \underbrace{\int a r' e^{-2r'/a}\, dr'}_{I_2}$$

I_2 también va por partes.

$$u = r' \qquad v = -\frac{a}{2} e^{-2r'/a}$$

$$du = dr' \qquad dv = e^{-2r'/a}\, dr'$$

$$I_2 = uv - \int v\,du$$

$$I_2 = -\frac{a}{2} r' e^{-2r'/a} - \int_0^{r'} -\frac{a}{2} e^{-2r'/a}\,dr'$$

$$I_2 = -\frac{a}{2} r' e^{-2r'/a} + \frac{a}{2}\int_0^{r'} e^{-2r'/a}\,dr$$

$$I_2 = -\frac{a}{2} r' e^{-2r'/a} + \frac{a}{2}\cdot -\frac{a}{2} e^{-2r'/a}$$

$$I_2 = \left[-\frac{a}{2} r' e^{-2r'/a} - \frac{a^2}{4} e^{-2r'/a} \right]_0^r$$

$$I_2 = -\frac{a}{2} r e^{-2r/a} - \frac{a^2}{4} e^{-2r/a} + \frac{a^2}{4}.$$

Entonces $I_1 = -\frac{a}{2} r^2 e^{-2r/a} + a \cdot I_2$

$$I_1 = -\frac{a}{2} r^2 e^{-2r/a} - \frac{a^2}{2} r e^{-2r/a} - \frac{a^3}{4} e^{-2r/a} + \frac{a^3}{4}.$$

$$I_1 = \frac{a}{2}\left(-r^2 e^{-2r/a} - a r e^{-2r/a} - \frac{a^2}{2} e^{-2r/a} + \frac{a^2}{2} \right)$$

$$Q_{enc}(r) = \frac{4q}{a^3} \cdot I_1$$

$$Q_{enc}(r) = \frac{2q}{a^2}\left(-r^2 e^{-2r/a} - a r e^{-2r/a} - \frac{a^2}{2} e^{-2r/a} + \frac{a^2}{2}\right)$$

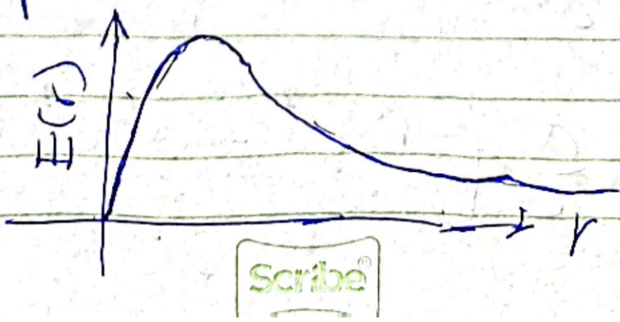
$$Q_{enc}(r) = q\left(-\frac{2}{a^2} r^2 e - \frac{2r}{a} e - e + 1\right)$$

$$Q_{enc}(r) = q\left(1 - e^{-2r/a}\left(1 + \frac{2r}{a} + \frac{2r^2}{a^2}\right)\right)$$

$$\bar{E}(r) = \frac{1}{4\pi\varepsilon_0} \frac{Q_{enc}}{r^2}$$

$$E(r) = \frac{q}{4\pi\varepsilon_0 r^2}\left(1 - e^{-2r/a}\left(1 + 2r/a + 2r^2/a^2\right)\right)$$ //

(b) El gráfico de $\bar{E}(r)$ se ve así



Problema #4

$A_{esf} = 4\pi r^2$

$V_{esf} = \frac{4}{3}\pi r^3$

De Gauss:

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\varepsilon_0}$$

Para una esfera gaussiana:

$$E \cdot (4\pi r^2) = \frac{Q_{enc}}{\varepsilon_0}$$

$$\Leftrightarrow E = \frac{1}{4\pi\varepsilon_0 r^2} \cdot Q_{enc}$$

Zona 1: $0 < r' < R_1$

$$\vec{E}_1 = \vec{0}$$

Zona 2: $R_1 < r' < R_2$

$$\rho_2 = \frac{Q_2}{V_2} = \frac{-3q}{\frac{4}{3}\pi(R_2^3 - R_1^3)}$$

$$\rho_2 = \frac{Q_2'}{V_2'} \rightarrow Q_2' = \rho_2 V_2'$$

$$\downarrow$$

$$Q_2' = \frac{Q}{V_2} \cdot V_2'$$

$$Q_2' = \frac{-3q}{\cancel{\frac{4}{3}}\pi (R_2^3 - R_1^3)} \cdot \cancel{\frac{4}{3}}\pi (r'^3 - R_1^3)$$

$$Q_2' = -3q \cdot \frac{(r'^3 - R_1^3)}{(R_2^3 - R_1^3)}$$

Ahora hay que sumarle la carga de zona 1.

$$Q_{enc}^{(2)} = Q_{z1} + Q_2'$$

$$Q_{enc}^{(2)} = q - 3q \frac{(r'^3 - R_1^3)}{(R_2^3 - R_1^3)}$$

Entonces

$$\vec{E}_2(r') = \frac{1}{4\pi\varepsilon_0 r'^2} \cdot \left(q - 3q \frac{(r'^3 - R_1^3)}{(R_2^3 - R_1^3)} \right).$$

Zona 3: $R_2 < r' < R_3$

$$q_{enc}^{(3)} = \frac{q}{8} - 3\frac{q}{8} = -2\frac{q}{8}$$

$$E_3(r') = \frac{1}{4\pi\epsilon_0 r'^2} \cdot \left(-2\frac{q}{8}\right)$$

Zona 4: $R_3 < r' < R_4$

$$\rho_4 = \frac{Q_4}{V_4} = \frac{2\frac{q}{8}}{\frac{4}{3}\pi\left(R_4^3 - R_3^3\right)}$$

$$\rho_4 = \frac{Q_4'}{V_4'} \rightarrow Q_4' = \rho_4 V_4'$$

$$\downarrow$$

$$Q_4' = \frac{Q_4}{V_4} \cdot V_4'$$

$$Q_4' = \frac{2\frac{q}{8}}{\cancel{\frac{4}{3}\pi}\left(R_4^3 - R_3^3\right)} \cdot \cancel{\frac{4}{3}\pi}\left(r'^3 - R_3^3\right)$$

$$Q_4' = 2\frac{q}{8} \cdot \frac{\left(r'^3 - R_3^3\right)}{\left(R_4^3 - R_3^3\right)}$$

Ahora hay que sumarle las cargas de zona 1, zona 2 y zona 3.

$$Q_{enc}^{(4)} = \frac{q}{8} + \left(-3\frac{q}{8}\right) + \left(-4\frac{q}{8}\right) + Q_4'$$

$$Q_{enc}^{(4)} = -6\frac{q}{8} + 2\frac{q}{8}\,\frac{(r'^3 - R_3^3)}{(R_4^3 - R_3^3)}$$

Entonces

$$E_4(r') = \frac{1}{4\pi\varepsilon_0 r'^2}\cdot\left(-6\frac{q}{8} + 2\frac{q}{8}\,\frac{(r'^3 - R_3^3)}{(R_4^3 - R_3^3)}\right).$$

Zona externa: $r' > R_4$

$$Q_{enc} = Q_{z1} + Q_{z2} + Q_{z3} + Q_{z4}.$$

$$Q_{enc} = \frac{q}{8} - 3\frac{q}{8} - 4\frac{q}{8} + 2\frac{q}{8} = -4\frac{q}{8}$$

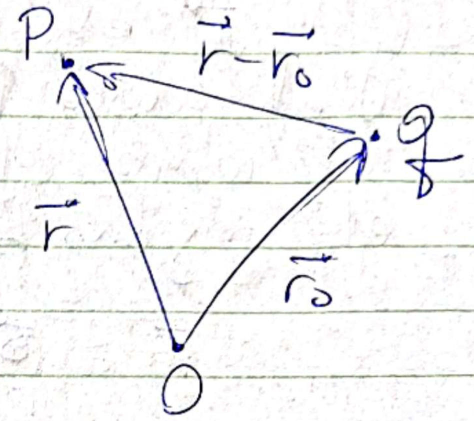
$$E(r') = \frac{1}{\cancel{4}\pi\varepsilon_0 r'^2}\cdot\left(-\cancel{4}\frac{q}{8}\right)$$

$$= \frac{-\frac{q}{8}}{\pi\varepsilon_0 r'^2}.$$

Problema # 5.

(a) Carga puntual:

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\varepsilon_0} \frac{q}{|\vec{r}-\vec{r}_0|^3} (\vec{r}-\vec{r}_0)$$



Varias cargas:

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\varepsilon_0} \sum_{i=1}^{N} \frac{q_i}{|\vec{r}-\vec{r}_i|^3} \cdot (\vec{r}-\vec{r}_i)$$

Para una distribución volumétrica:

$$d\vec{E}(\vec{r}) = \frac{1}{4\pi\varepsilon_0} \frac{dq'}{|\vec{r}-\vec{r}'|^3} (\vec{r}-\vec{r}')$$

$$\rho = \frac{q}{V}$$

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\varepsilon_0} \int_V \frac{\rho \, dV'}{|\vec{r}-\vec{r}'|^3} \cdot (\vec{r}-\vec{r}')$$

$$\rho = \frac{dq'}{dV'}$$

$$\rightarrow dq' = \rho \, dV'$$

Donde el prima (') indica elemento diferencial

En esa expresión

$$\rho = \rho(\vec{r}\,')$$

y $\vec{r}$ es el radio vector del origen al punto P (donde queremos determinar el campo Eléctrico). Pero P es, justamente el origen. Entonces

$$\vec{r} = \vec{0}.$$

Por lo que:

$$\vec{E}(0) = \frac{1}{4\pi\varepsilon_0}\int_V \frac{\rho(\vec{r}\,')\,dV'}{|\vec{r}\,'|^3}(-\vec{r}\,')$$

Usando esféricos:

$$x = r \sin\theta \cos\phi \qquad r: 0 \leq r$$

$$y = r \sin\theta \sin\phi \qquad \theta: [0, \pi]$$

$$z = r \cos\theta \qquad \phi: [0, 2\pi[$$

$$dV = r^2 \sin\theta \, dr\, d\theta\, d\phi$$

Entonces

$$\vec{E}(0) = \frac{-1}{4\pi\varepsilon_0}\int_V \frac{\rho_0 \frac{r}{R_2^2}\,(\vec{r}\,)}{\cancel{r^3}} \cdot \cancel{r^2}\sin\theta \, dr\, d\theta\, d\phi$$

Con $r: R_1 \rightarrow R_2$

$\theta:$ $\pi/2 \rightarrow \pi$

$\phi:$ $3\pi/2 \rightarrow 2\pi$

$$\vec{E}(0) = \frac{-\rho_0}{4\pi\varepsilon_0 R_2^2}\int_{R_1}^{R_2}\int_{\frac{\pi}{2}}^{\pi}\int_{\frac{3\pi}{2}}^{2\pi} \vec{r}\,\sin\theta \, d\phi\, d\theta\, dr$$

$$\vec{r} = (x, y, z)$$

$$\vec{r} = r(\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$$

$$\vec{E}(0)_x = \frac{-\rho_0}{4\pi\varepsilon_0 R_2^2}\int_{R_1}^{R_2} r\, dr \int_{\frac{\pi}{2}}^{\pi}\sin^2\theta\, d\theta \int_{\frac{3\pi}{2}}^{2\pi}\cos\phi\, d\phi.$$

$\vec{E}(0)_y = \vec{E}(0)_x$, por simetría.

$$E(0)_z = \frac{-\rho_0}{4\pi\varepsilon_0 R_2^2}\int_{R_1}^{R_2} r\, dr \int_{\frac{\pi}{2}}^{\pi}\sin\theta\, d\theta * \cos\theta \int_{\frac{3\pi}{2}}^{2\pi} d\phi.$$

$$E(0)_x = \frac{-\rho_0}{4\pi\varepsilon_0 R_2^2}\left(\frac{R_2^2-R_1^2}{2}\right)\left(\frac{\pi}{4}\right)(1)$$

$$E(0)_x = \frac{-\rho_0 (R_2^2-R_1^2)}{8\pi\varepsilon_0 R_2^2}\cdot\frac{\pi}{4} = E(0)_y$$

$$E(0)_z = \frac{-\rho_0}{4\pi\varepsilon_0 R_2^2}\left(\frac{R_2^2-R_1^2}{2}\right)\left(\frac{-1}{2}\right)\left(\frac{\pi}{2}\right)$$

$$E(0)_z = \frac{+\rho_0 (R_2^2-R_1^2)}{8\pi\varepsilon_0 R_2^2}\cdot\frac{\pi}{4}.$$

Entonces

$$\vec{E}(0) = \frac{\rho_0 (R_2^2-R_1^2)}{8\cancel{\pi}\varepsilon_0 R_2^2}\,\frac{\cancel{\pi}}{4}\,(-1,1,1)$$

$$\boxed{\vec{E}(0) = \frac{\rho_0 (R_2^2-R_1^2)}{32\,\varepsilon_0 R_2^2}\,(-1,1,1)}$$

(b) $\vec{E} = \dfrac{\vec{F}}{q}$

$$\vec{F} = q\,\vec{E}$$

$$\vec{F} = \frac{-3q\,\rho_0(R_2^2 - R_1^2)}{3\sqrt{2}\,\varepsilon_0 R_2^2}\,(-1,1,1)$$

$$\vec{F} = \frac{q\,\rho_0(R_2^2 - R_1^2)}{3\sqrt{2}\,\varepsilon_0 R_2^2}\,(1,-1,-1)$$